$\forall \varepsilon > 0, \exists\ \delta > 0$ such that $|f(x) - L| < \varepsilon$ whenever $0 < |x - x_0| < \delta$.

This means whenever $x \neq x_0$ and is within δ distance from x_0, then $f(x)$ is within ε distance from L. Following figures explain the interplay between ε and δ.

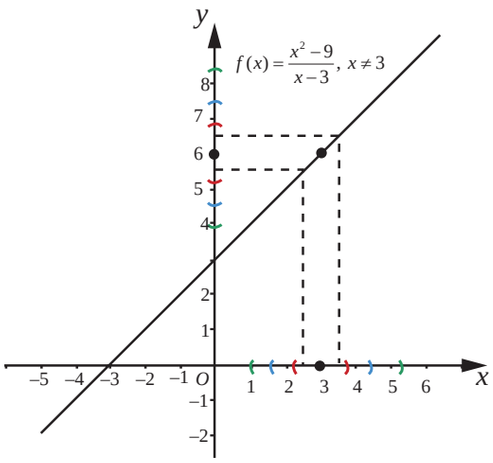


Fig. 8.7

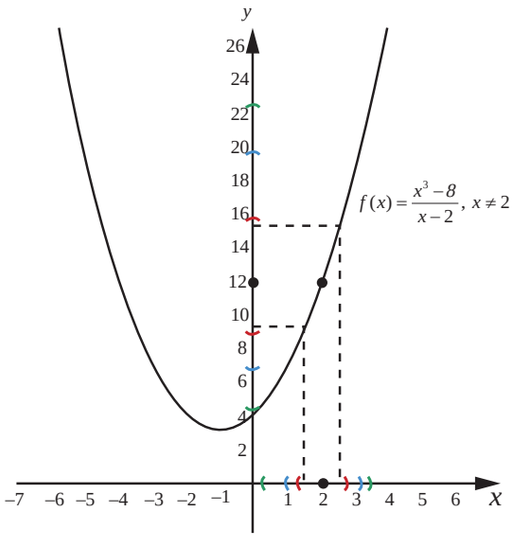


Fig. 8.8

We also know, from XI Std, that a function f defined in the neighbourhood of x_0 except possibly at x_0 has a limit at x_0 if the following hold :

(1) $\lim\limits_{x \to x_0^+} f(x) = L_1$ (right hand limit) exists

(2) $\lim\limits_{x \to x_0^-} f(x) = L_2$ (left hand limit) exists

(3) $L_1 = L_2$.

Let $f(x_0) = L$ (say). Then the function f is continuous at x= x_0 if $L = L_1 = L_2$ .Note that in the limit and continuity of a single variable functions, neighbourhoods play an important role. In this case a neighbourhood of a point $x_0 \in \mathbb{R}$ looks like $(x_0 - r, x_0 + r)$, where $r > 0$. In order to develop limit and continuity of functions of two variables, we need to define neighbourhood of a point $(u, v) \in \mathbb{R}^2$. So, for $(u, v) \in \mathbb{R}^2$ and $r > 0$, a r-neighbourhood of the point (u, v) is the set

$$B_r((u,v)) = \{(x,y) \in \mathbb{R}^2 \mid (x-u)^2 + (y-v)^2 < r^2\}.$$

So a r-neighbourhood of a point (u, v) is an open disc with centre (u, v) and radius $r > 0$. If the centre is removed from the neighbourhood, then it is called a **deleted neighbourhood**.

8.4 Limit and Continuity of Functions of Two Variables

Definition 8.6 (Limit of a Function)

Suppose that $A = \{(x,y) \mid a < x < b, c < y < d\} \subset \mathbb{R}^2, F : A \to \mathbb{R}$. We say that F has a limit L at (u, v) if the following hold :

For every neighboourhood $(L - \varepsilon, L + \varepsilon), \varepsilon > 0$, of L, there exists a δ-neighbourhood $B_\delta((u,v)) \subset A$ of (u, v) such that $(x,y) \in B_\delta((u,v)) \setminus \{(u,v)\}, \delta > 0 \Rightarrow f(x) \in (L - \varepsilon, L + \varepsilon)$.

We denote this by $\lim\limits_{(x,y) \to (u,v)} F(x,y) = L$ if such a limit exists.

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